

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: DATA VISUALIZATION WITH POWER BI & TABLEAU

PRACTICAL - 9

Aim: Cleaning and Structuring Data for Analysis

- Identify and handle missing or inconsistent data
- Convert wide data into tall format
- Perform union of multiple files
- Execute cross-database joins
- Restructure data for better visualization
- Handle different levels of detail in datasets
- Apply advanced data cleaning techniques

A: Identify and Handle Missing or Inconsistent Data

Step 1: Load the Dataset

- Open Tableau Desktop.
- Under the Connect pane on the left, click Text file.
- Select and open Sem1_Scores.csv.
- In the Data Source preview table:
 - Observe missing entries (Null) in Math, Physics, and Chemistry.

The screenshot shows the Tableau Desktop interface with the 'Sem1_Scores.csv' data source loaded. The left pane shows the 'Connections' and 'Files' sections. The 'Files' section lists several CSV files, including 'Sem1_Scores.csv'. The central area displays a preview of the data source, showing a table with columns: Student ID, Full Name, Dept Info, Math, and Physics. The 'Math' column contains a null value for the student S207. The bottom status bar shows the system date and time as 19:16 on 23-08-2026.

Student ID	Full Name	Dept Info	Math	Physics
S206	Riya Kapoor	CSE-01	87	0.900000
S207	Manav Shah	ECE-02	null	0.740000
S208	Tanya Rao	IT-03	93	0.960000
S209	Dev Malhotra	CSE-01	71	0.650000

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Step 2: Clean Inconsistent String Fields (Convert Text Percentages to Numeric Measures)

1. Navigate to Sheet 1 (bottom-left tab).

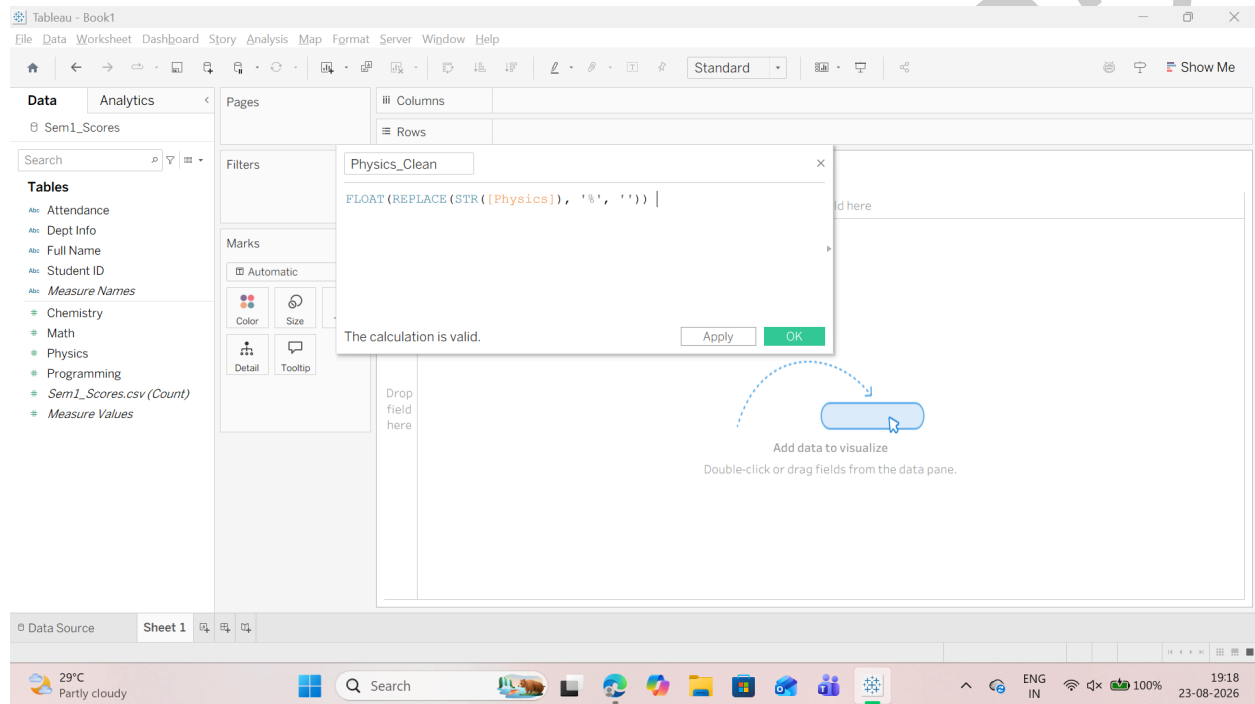
2. Right-click an empty space in the Data Pane (left sidebar) --> select Create Calculated Field....

3. Create the cleaned measure for Physics:

o Field Name: Physics_Clean

o Tableau Formula:

`FLOAT(REPLACE(STR([Physics]), '%', ''))`



4. Create the cleaned measure for Attendance: o Right-click the Data Pane --> Create Calculated Field....

o Field Name: Attendance_Clean

o Tableau Formula:

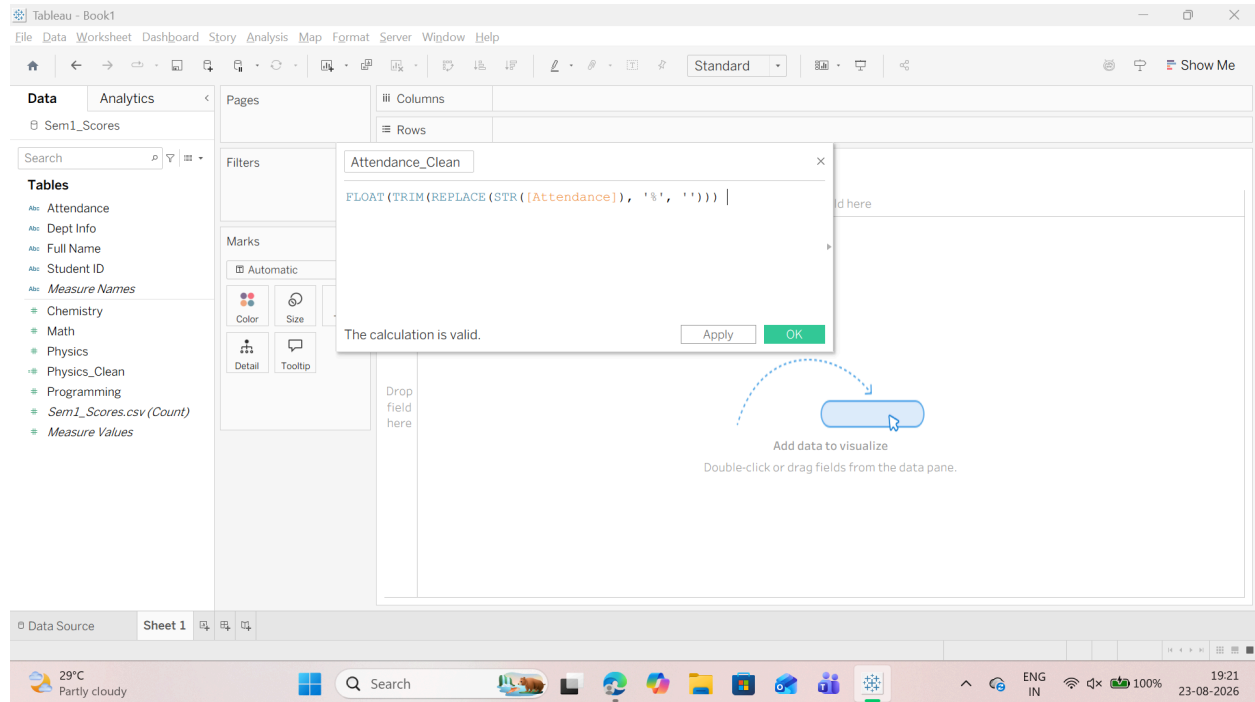
Code snippet

`FLOAT(TRIM(REPLACE(STR([Attendance]), '%', '')))`

o Click OK.

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Step 3: Handle Missing (Null) Numeric Values via Imputation

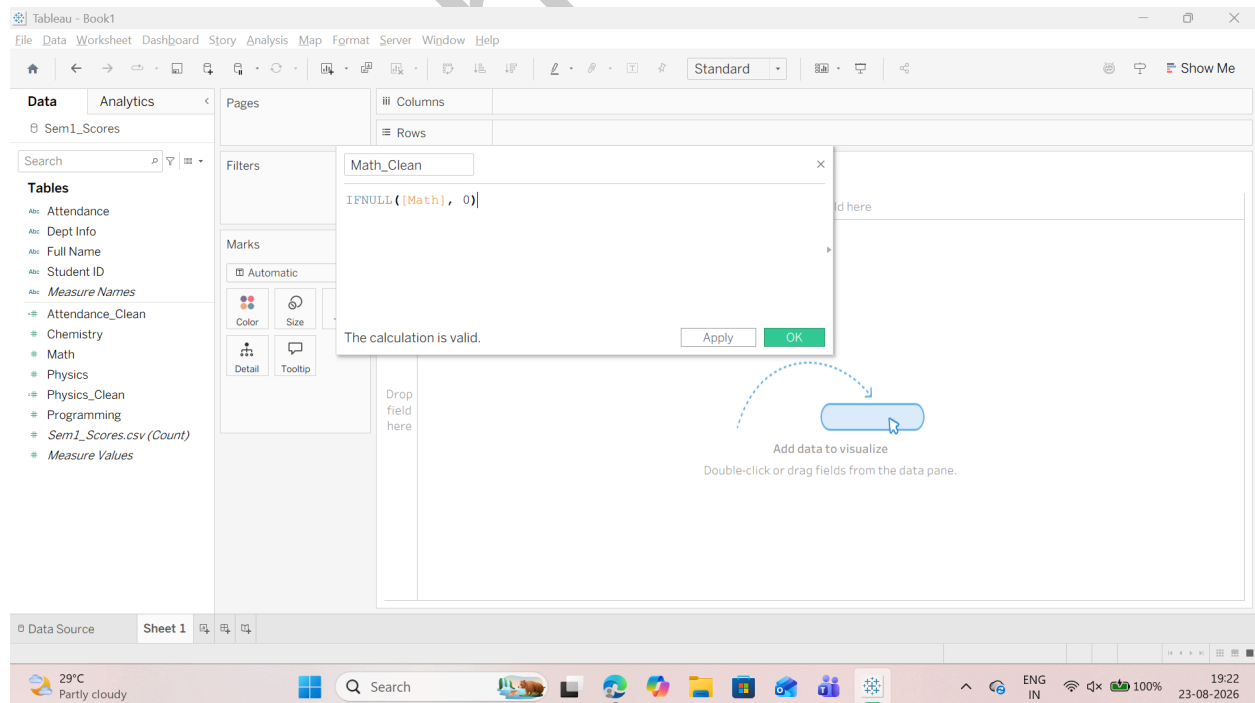
1. Handle missing entries in Math:

o Right-click the Data Pane --> Create Calculated Field....

o Field Name: Math_Clean

o Tableau Formula:

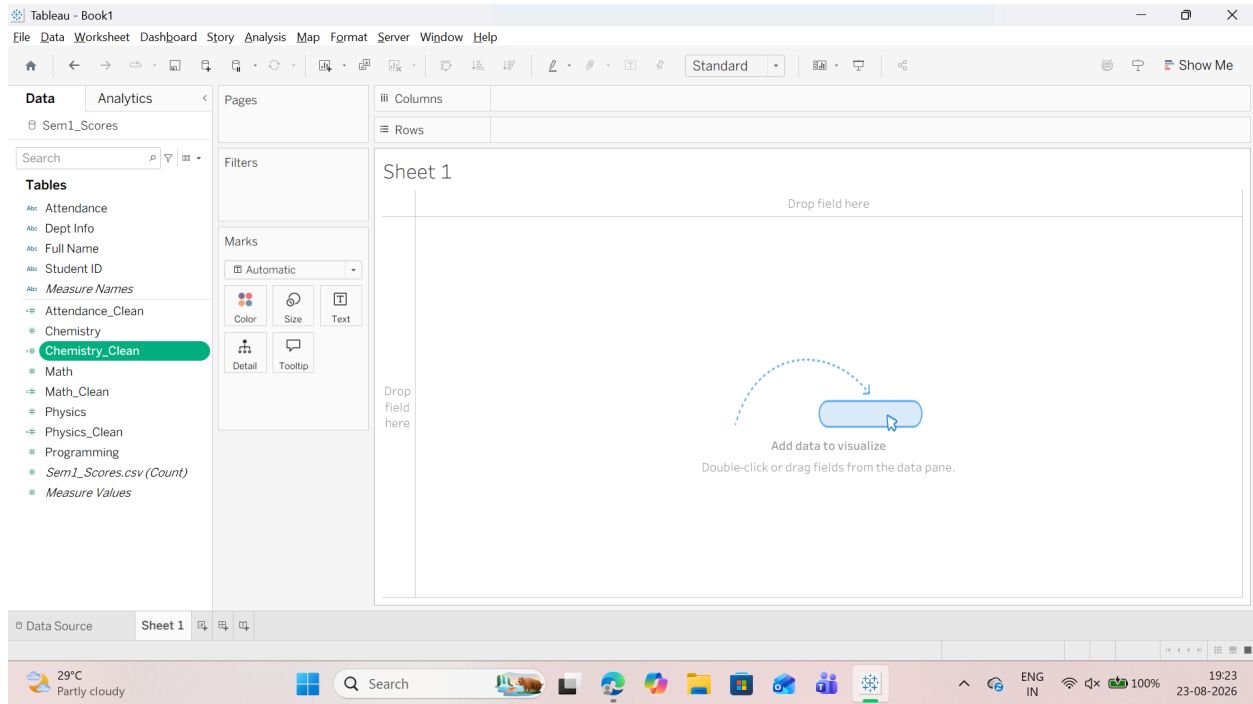
`IFNULL([Math], 0)`



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2.Right-click Data Pane → Create Calculated Field → Name: Chemistry_Clean → Formula: IFNULL([Chemistry], 0) → Click OK.

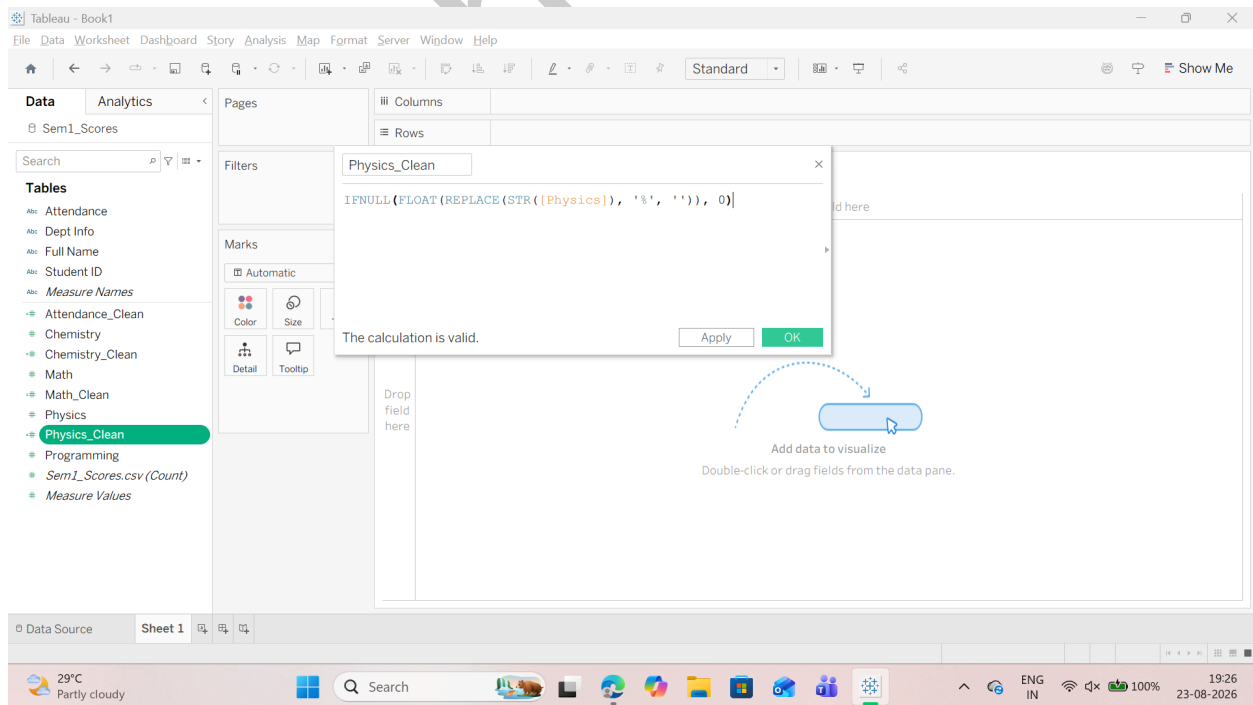


Step 4: Impute Missing Values in Physics (Optional Handling for Student S105)

1.S105 contains a Null value in Physics. Combine string stripping with null handling:

o Edit Physics_Clean (right-click Physics_Clean --> Edit...):

IFNULL(FLOAT(REPLACE(STR([Physics]), '%', '')), 0)



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B: Convert Wide Data into Tall Format

Step 1: Open the Data Source Grid

1. In Tableau, navigate to the Data Source tab at the bottom-left corner.
2. Locate the four horizontal subject score columns:
 - o Math o Chemistry
 - o Programming o Physics

Step 2: Multi-Select and Pivot Columns

1. Click the header of the Math column.
2. Hold down Ctrl (or Cmd on Mac) and click Physics, Chemistry, and Programming so all four columns are highlighted in blue.
3. Right-click on any of the highlighted column headers.
4. Select Pivot.

The screenshot shows the Tableau Desktop interface with the 'Sem1_Scores' data source loaded. The 'Data Source' tab is active, displaying a grid of fields. The 'Math', 'Physics', 'Chemistry', and 'Programming' columns are selected and pivoted. The resulting table has columns for Student ID, Full Name, Dept Info, and four subject scores (Math, Physics, Chemistry, Programming), followed by Attendance and two calculated fields (Physics_Clean and Calculated Attend).

Student ID	Full Name	Dept Info	Math	Physics	Chemistry	Programming	Attendance	Physics_Clean	Calculated Attend
S201	Neel Joshi	CSE-01	91	0.870000	89	94	92 %	0.870000	
S202	Meera Shah	ECE-02	76	0.810000	84	79	86 %	0.810000	
S203	Karan Mehta	CSE-01	83	0.890000	78	91	88 %	0.890000	
S204	Isha Kulkarni	IT-03	96	0.930000	94	97	96 %	0.930000	
S205	Arjun Desai	MECH-04	58	null	62	68	70 %	0.000000	
S206	Riya Kapoor	CSE-01	87	0.900000	86	93	91 %	0.900000	
S207	Manav Shah	ECE-02	null	0.740000	77	82	75 %	0.740000	
S208	Tanya Rao	IT-03	93	0.960000	91	99	97 %	0.960000	

Step 3: Rename the Pivoted Fields Tableau replaces the four columns with two consolidated columns at the end of the table:

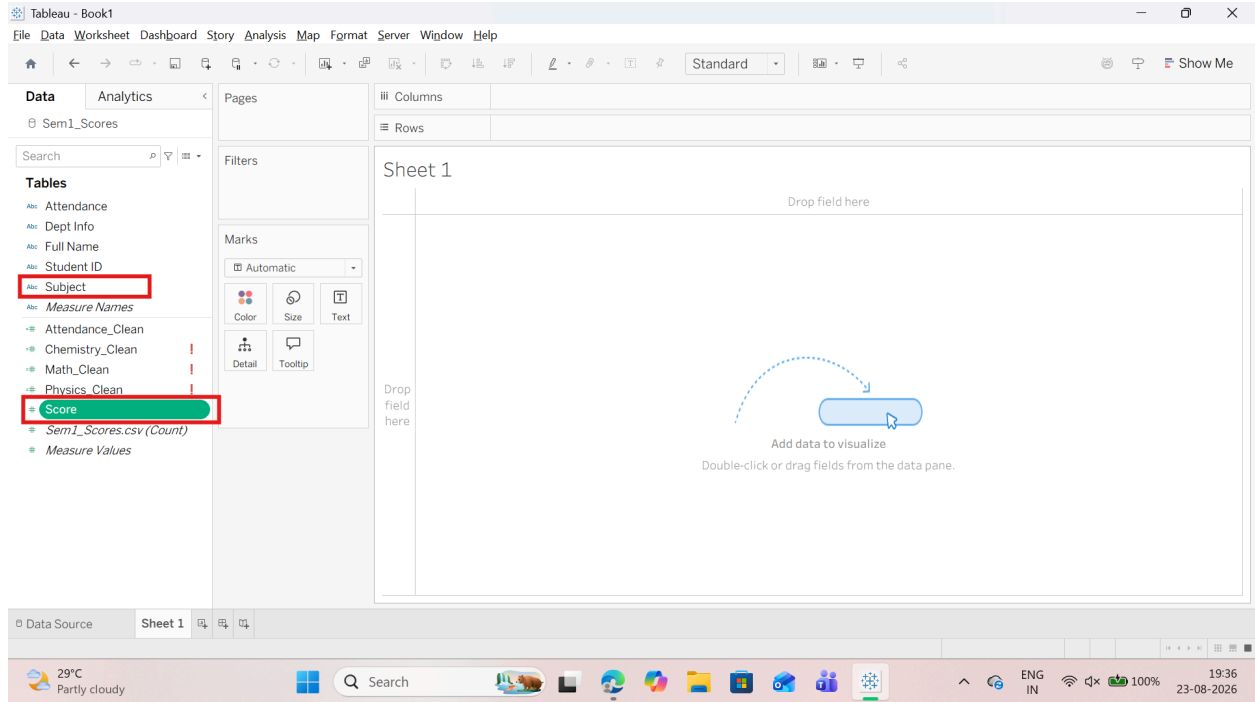
1. Double-click the header named Pivot Field Names and type: Subject
2. Double-click the header named Pivot Field Values and type: Score

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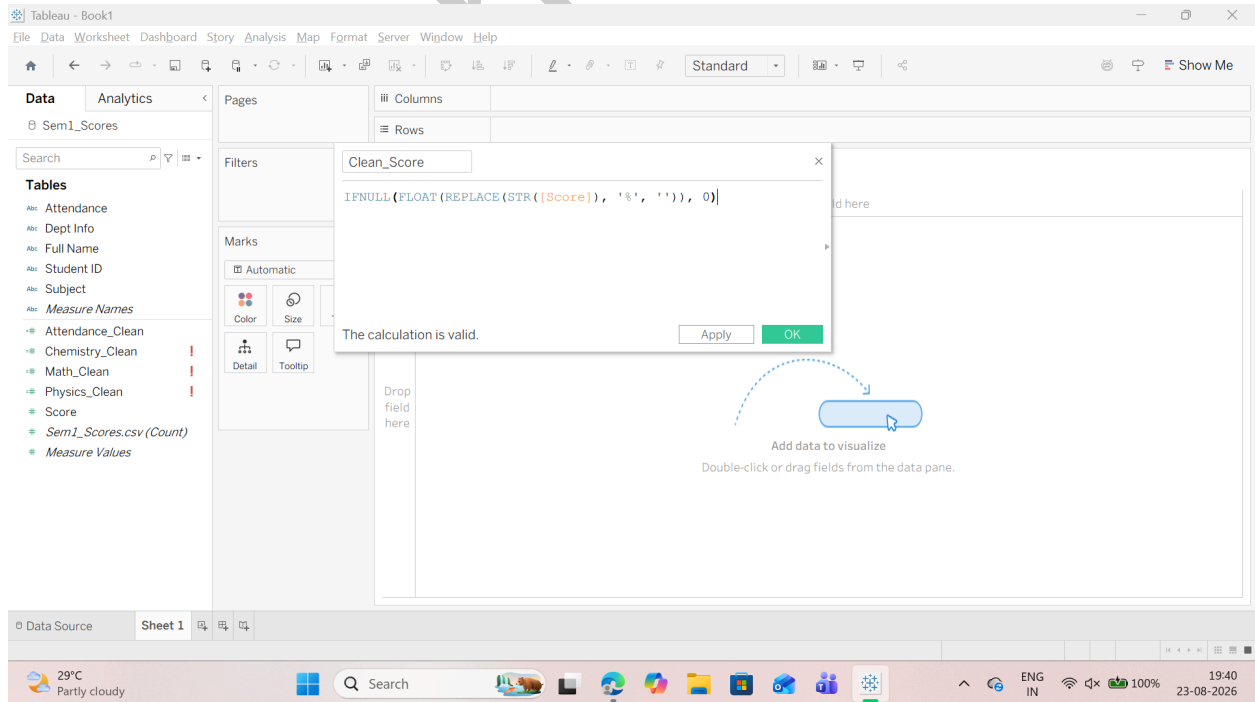
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Step 4: Clean and Format the Tall Score Field Because raw Physics contained % symbol, the new combined Score column defaults to text (Abc). Convert it to numeric measure:

1. Click the Abc icon directly above the Score column header.
2. Select Number (decimal) from the dropdown list.
3. Alternatively, create a calculated field inside Sheet 1 named Clean_Score:
`IFNULL(FLOAT(REPLACE(STR([Score]), '%', '')), 0)`



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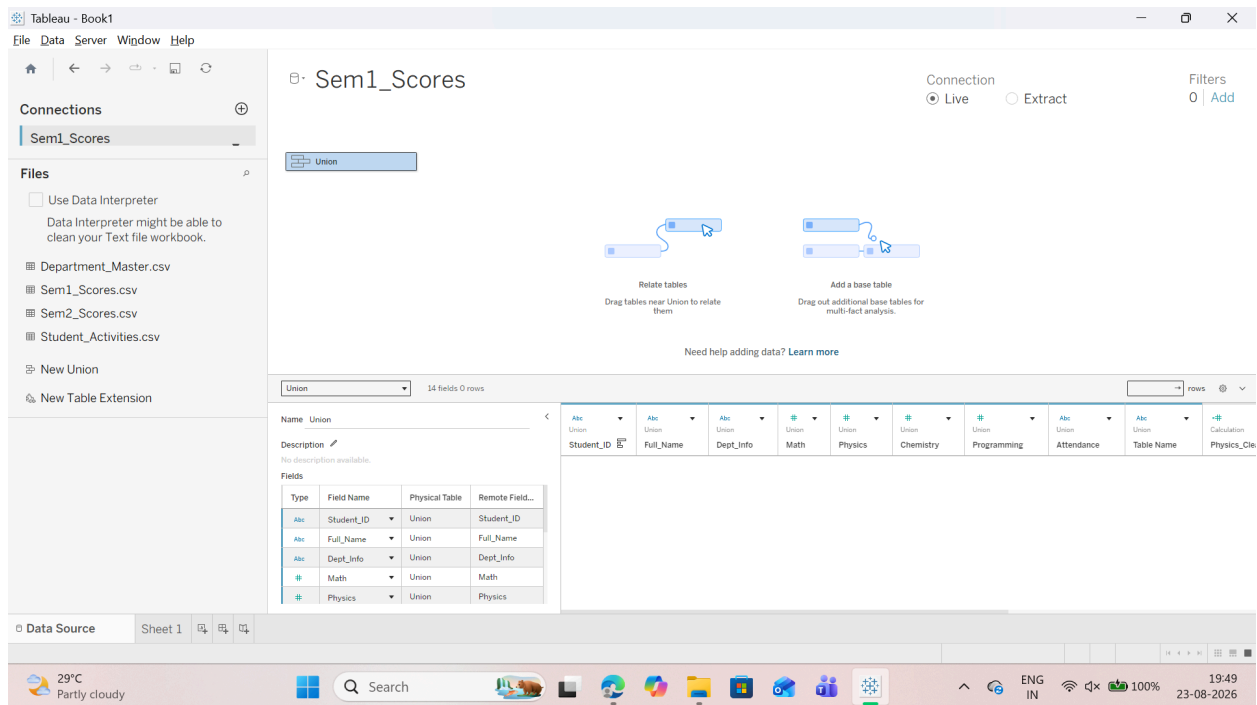
C: Perform Union of Multiple Files

Step 1: Open the Data Source Canvas

1. Open Tableau Desktop and go to the Data Source tab in the bottom-left corner.
2. If any single table is already on the canvas, right-click it and select Remove to start with a clean canvas.

Step 2: Create a Multi-File Union

1. Look at the left sidebar under the Files / Connections pane.
2. Drag the New Union option onto the empty canvas area.
3. In the pop-up Union dialog box:
 - o Drag Sem1_Scores.csv from the left panel into the box.
 - o Drag Sem2_Scores.csv from the left panel into the same box.
 - o Verify that both filenames are listed in the Union box.
4. Click Apply, then click OK.
5. The canvas table icon will now display with a plus symbol (+), representing the consolidated union.



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Step 3: Verify the Merged Data Grid

1. Review the data grid at the bottom of the screen.
2. Confirm the row count: the total number of rows increases from 10 to 20 rows (10 records for Semester 1 and 10 records for Semester 2).
3. Scroll to the far right of the grid to verify the generated system column:
 - o Table Name: Displays "Sem1_Scores.csv" for the first 10 rows and "Sem2_Scores.csv" for the remaining 10 rows.

The screenshot shows the Tableau interface with a Union of two CSV files. The data grid at the bottom displays 20 rows. The 'Table Name' column on the far right is highlighted with a red box, showing 'Sem1_Scores.csv' for the first 10 rows and 'Sem2_Scores.csv' for the next 10 rows.

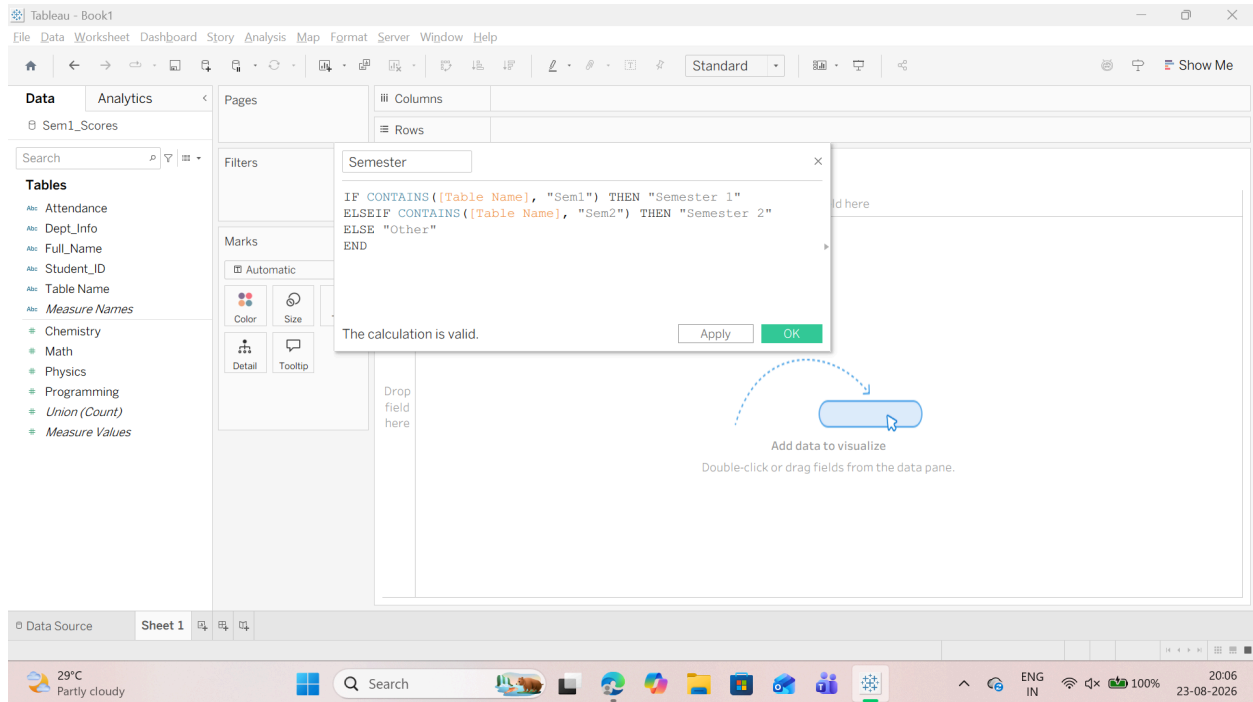
Student_ID	Full_Name	Dept_Info	Math	Physics	Chemistry	Programming	Attendance	Table Name
S206	Riya Kapoor	CSE-01	87	0.900000	86	93	91 %	Sem1_Scores.csv
S207	Manav Shah	ECE-02	null	0.740000	77	82	75 %	Sem1_Scores.csv
S208	Tanya Rao	IT-03	93	0.960000	91	99	97 %	Sem1_Scores.csv
S209	Dev Malhotra	CSE-01	71	0.650000	63	75	82 %	Sem1_Scores.csv
S210	Nisha Menon	ECE-02	85	0.880000	82	90	94 %	Sem1_Scores.csv
S201	Neel Joshi	CSE-01	84	0.890000	86	93	91 %	Sem2_Scores.csv
S202	Meera Shah	ECE-02	81	0.830000	85	90	89 %	Sem2_Scores.csv
S203	Karan Mehta	CSE-01	87	0.880000	82	94	86 %	Sem2_Scores.csv
S204	Isha Kulkarni	IT-03	92	0.950000	93	97	98 %	Sem2_Scores.csv

Step 4: Create a Clean Semester Identifier Dimension

1. Click on Sheet 1 at the bottom-left.
2. In the Data Pane (left sidebar), right-click in an empty space select Create Calculated Field....
3. Set the Field Name to: Semester
4. Enter the following formula:
Code snippet
IF CONTAINS([Table Name], "Sem1") THEN "Semester 1"
ELSEIF CONTAINS([Table Name], "Sem2") THEN "Semester 2"
ELSE "Other"
END

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D: Execute Cross-Database Joins

Step 1: Stay on the Top Canvas

Step 2: Add and Link Department_Master.csv

1. In the left sidebar under Files / Connections, find Department_Master.csv.
2. Drag Department_Master.csv onto the canvas and drop it to the right of your Union box.
3. A flexible line (relationship / join noodle) will connect the two boxes.
4. If the relationship dialog opens (or click the relationship line):
 - o Left Table Field (Union): Select Dept_Info
 - o Right Table Field (Department_Master): Select Dept_Code
 - o Operator: =

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Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv

New Union

New Table Extension

Sem1_Scores

Connection: Live Extract

Filters: 0 Add

Union

Department_Master.csv

Union — Department_Master.csv

Name: Department_Master.csv

How do relationships differ from joins? Learn more

Union Operator Department_Master.csv

Dept Code

Dept Name

HOD Name

Dept Target Average

Total Dept Capacity

CSE-01 Computer Science Dr. Mehta 87 125

ECE-02 Electronics & Communication Dr. Kapoor 80 95

IT-03 Information Technology Dr. Nair 84 70

MECH-04 Mechanical Engineering Dr. Kulkarni 72 65

Performance Options

Remove Relationship

Data Source Sheet 1

28°C Mostly cloudy

Search

ENG IN

100%

20:09 23-08-2026

Step 3: Add and Link Student_Activities.csv

1. From the left sidebar, drag Student_Activities.csv onto the canvas.
2. Drop it to the right of the Union box.
3. In the relationship dialog:
 - o Left Table Field (Union): Select Student_ID
 - o Right Table Field (Student_Activities): Select Student_ID

Tableau - Book1

File Data Server Window Help

Connections

Sem1_Scores

Files

Use Data Interpreter
Data Interpreter might be able to clean your Text file workbook.

Department_Master.csv
Sem1_Scores.csv
Sem2_Scores.csv
Student_Activities.csv

New Union

New Table Extension

Sem1_Scores

Connection: Live Extract

Filters: 0 Add

Union

Department_Master.csv

Student_Activities.csv

Union — Student_Activities.csv

Name: Student_Activities.csv

How do relationships differ from joins? Learn more

Union Operator Student_Activities.csv

Student ID (Student Activ)

Club Membership

Projects Completed

Certifications

S201 AI Club 4 3

S202 Robotics Club 3 2

S203 Coding Club 3 1

S204 Literary Society 5 4

S205 Sports Club 2 1

S206 AI Club 4 3

S207 Robotics Club 3 2

S208 Literary Society 5 4

S209 Sports Club 2 2

Performance Options

Remove Relationship

Data Source Sheet 1

28°C Mostly cloudy

Search

ENG IN

100%

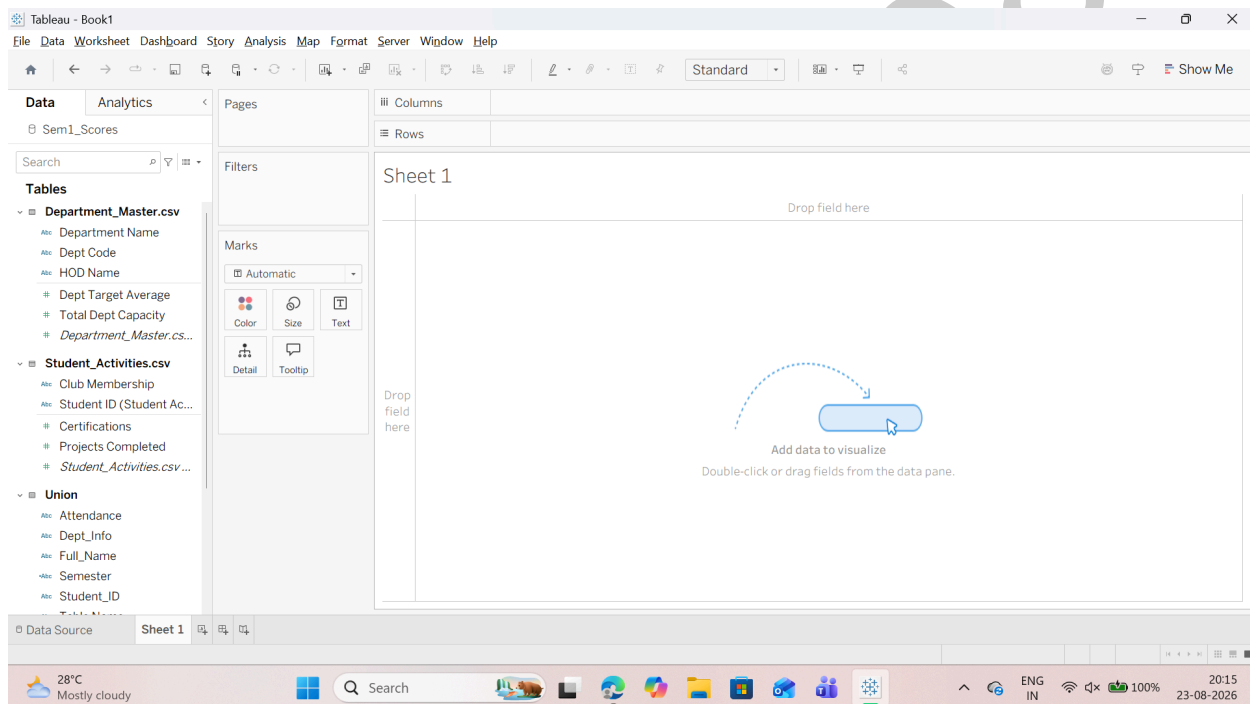
20:11 23-08-2026

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Step 4: Verify in Sheet 1

1. Click on Sheet 1 at the bottom-left.
2. In the Data Pane on the far left, you will now see all three tables organized with their respective fields:
 - o Union: Student_ID, Full_Name, Dept_Info, Math, Physics, Chemistry, Programming, Attendance, Table Name
 - o Department_Master: Dept_Code, Department_Name, HOD_Name, Dept_Target_Average, Total_Dept_Capacity
 - o Student_Activities: Club_Membership, Projects_Completed, Certifications



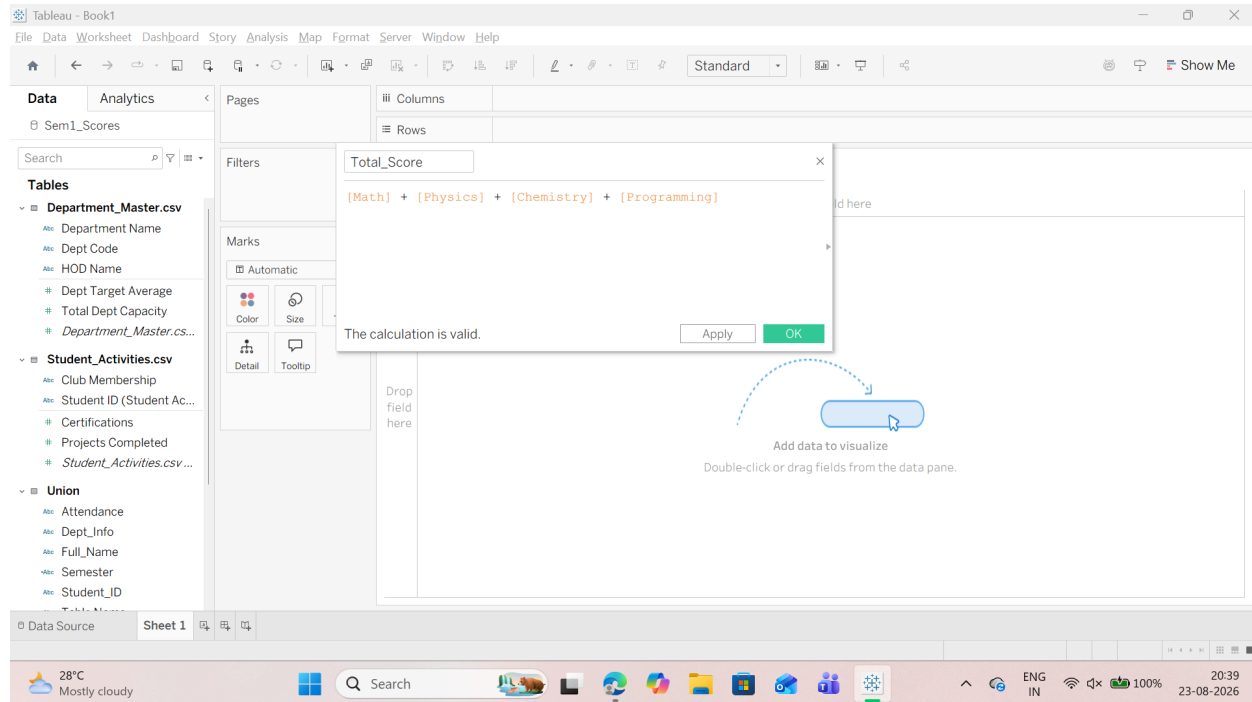
E: Restructure Data for Better Visualization

Step 1: Calculate Total Marks and Scaled Percentage

1. Navigate to Sheet 1 in Tableau.
2. In the Data Pane on the left, right-click an empty space --> select Create Calculated Field....
3. Create the student's total marks field:
 - o Field Name: Total_Score
 - o Tableau Formula:
$$[\text{Math_Clean}] + [\text{Physics_Clean}] + [\text{Chemistry_Clean}] + [\text{Programming}]$$

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4. Create the scaled overall percentage field:

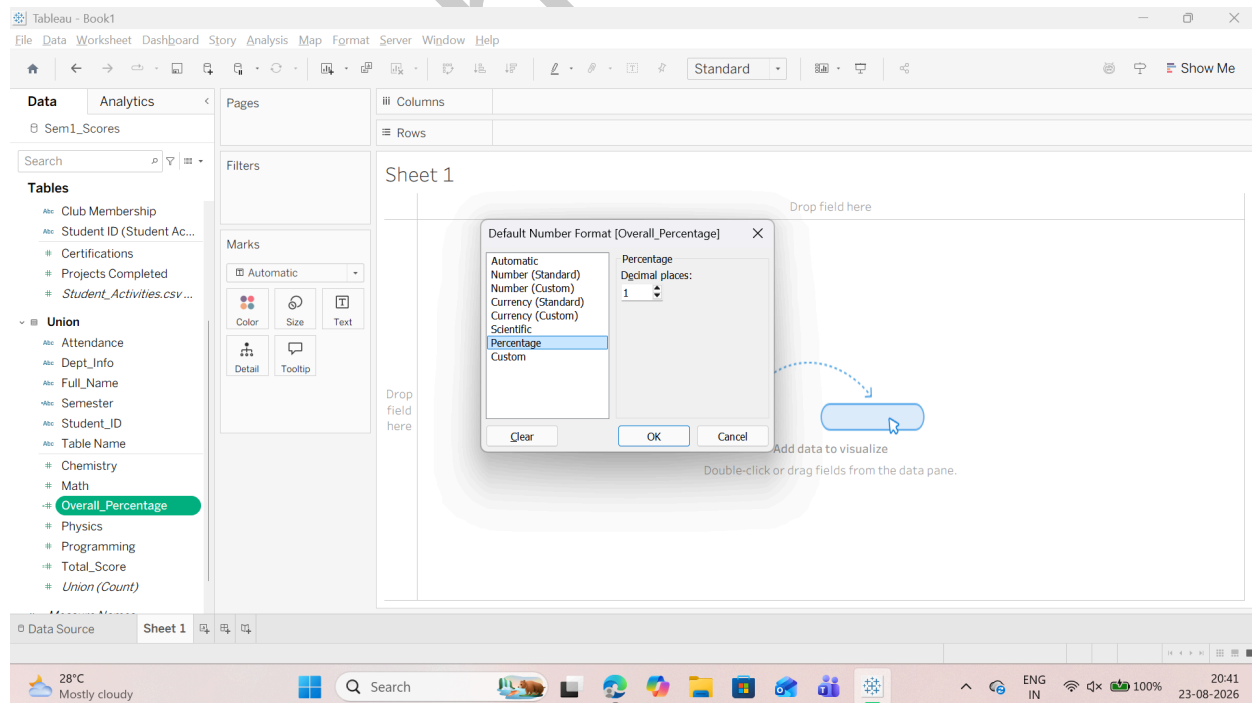
- o Right-click in the Data Pane --> select Create Calculated Field....

- o Field Name: Overall_Percentage o Tableau Formula: $[Total_Score] / 400.0$

5. Set the default percentage display formatting:

- o In Data Pane, right-click Overall_Percentage >Default Properties>Number Format....

- o Select Percentage and set decimal places to 1 (e.g., 85.5%).



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Step 2: Create Categorical Performance Bands (Grading Tiers)

1. Right-click in the Data Pane -->select Create Calculated Field....

2. Create the academic grading tier field:

o Field Name: Performance_Band

o Tableau Formula:

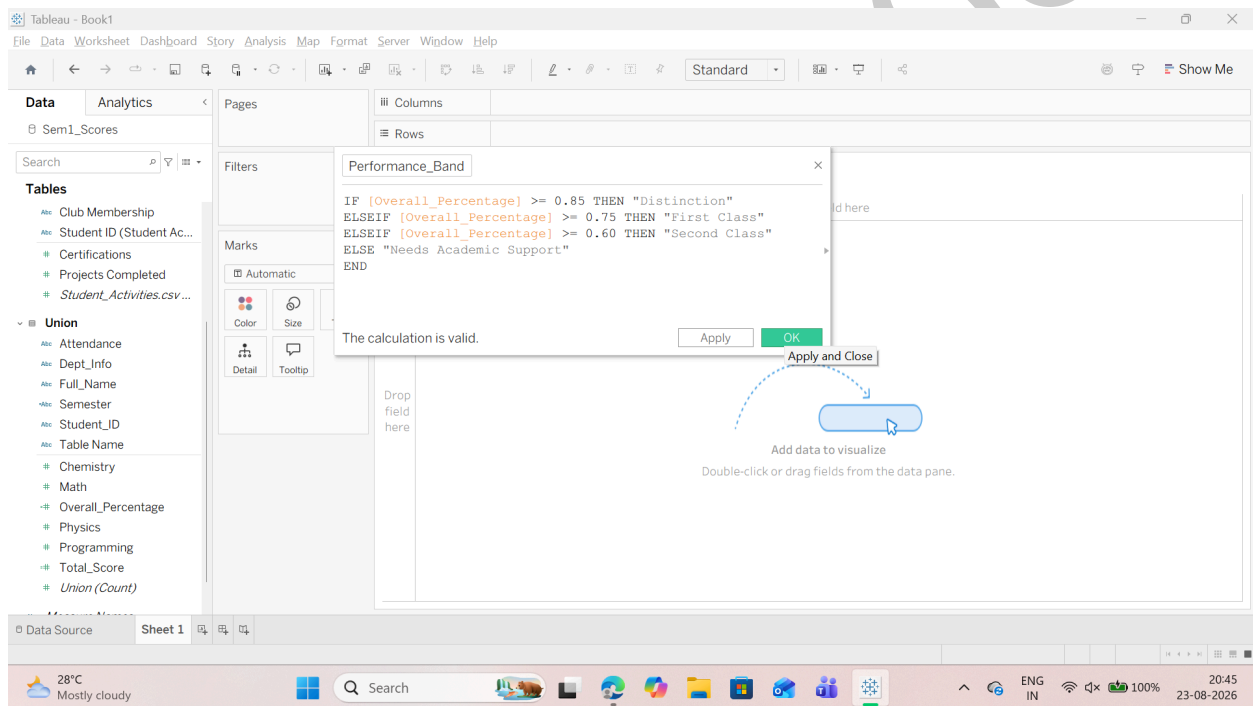
```
IF [Overall_Percentage] >= 0.85 THEN "Distinction"
```

```
ELSEIF [Overall_Percentage] >= 0.75 THEN "First Class"
```

```
ELSEIF [Overall_Percentage] >= 0.60 THEN "Second Class"
```

```
ELSE "Needs Academic Support"
```

```
END
```



Step 3: Create an Attendance Health Flag

1. Right-click in the Data Pane -->select Create Calculated Field....

2. Create the attendance compliance indicator:

o Field Name: Attendance_Status

o Tableau Formula:

```
IF [Attendance_Clean] >= 75 THEN "Eligible (Good Standing)"
```

```
ELSE "Shortage Warning (<75%)"
```

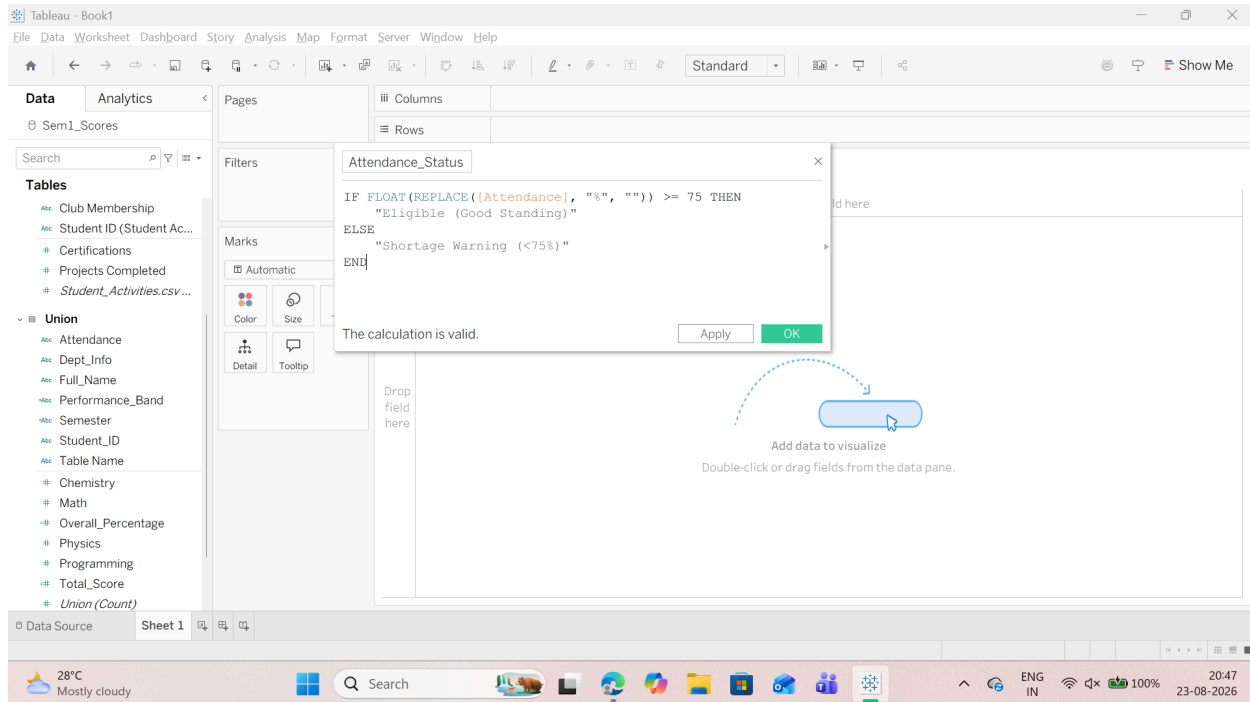
```
END
```

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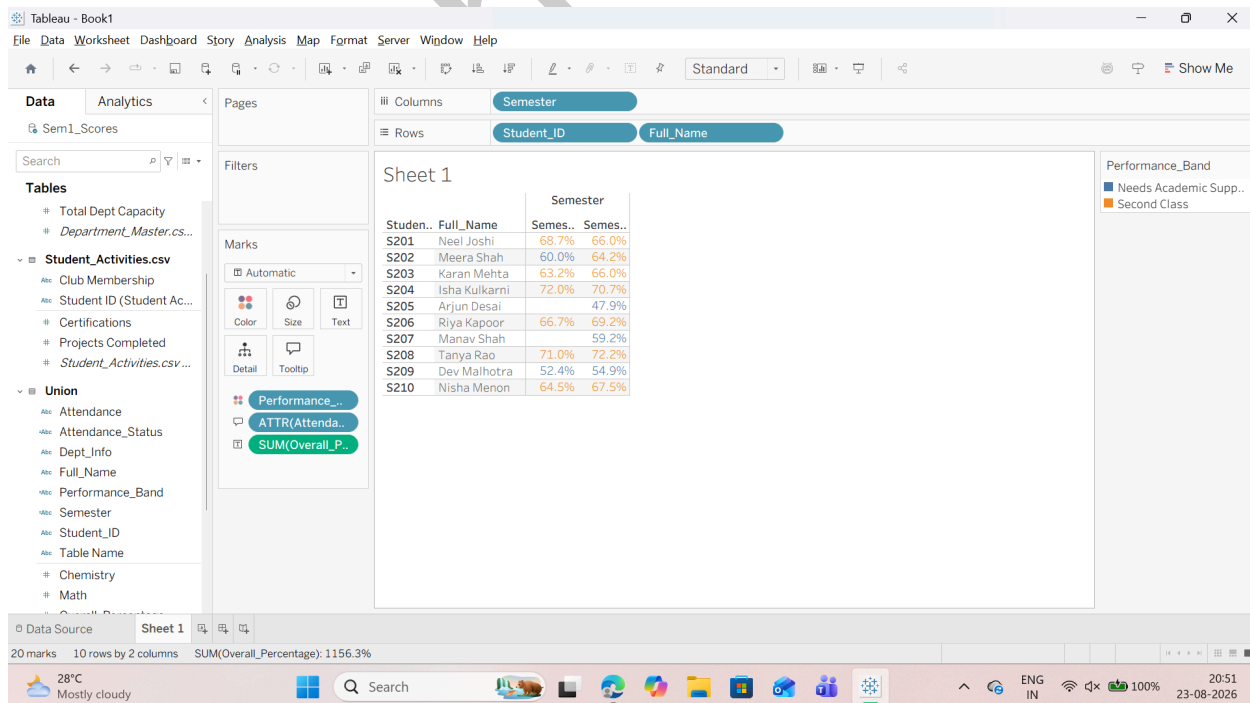
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Step 4: Build the Student Performance Scorecard View

1. Drag Student_ID and Full_Name onto the Rows shelf.
2. Drag Semester onto the Columns shelf.
3. Drag Overall_Percentage onto the Text tile on the Marks card.
4. Drag Performance_Band onto the Color tile on the Marks card.
5. Drag Attendance_Status onto the Tooltip (or Detail) tile on the Marks card.



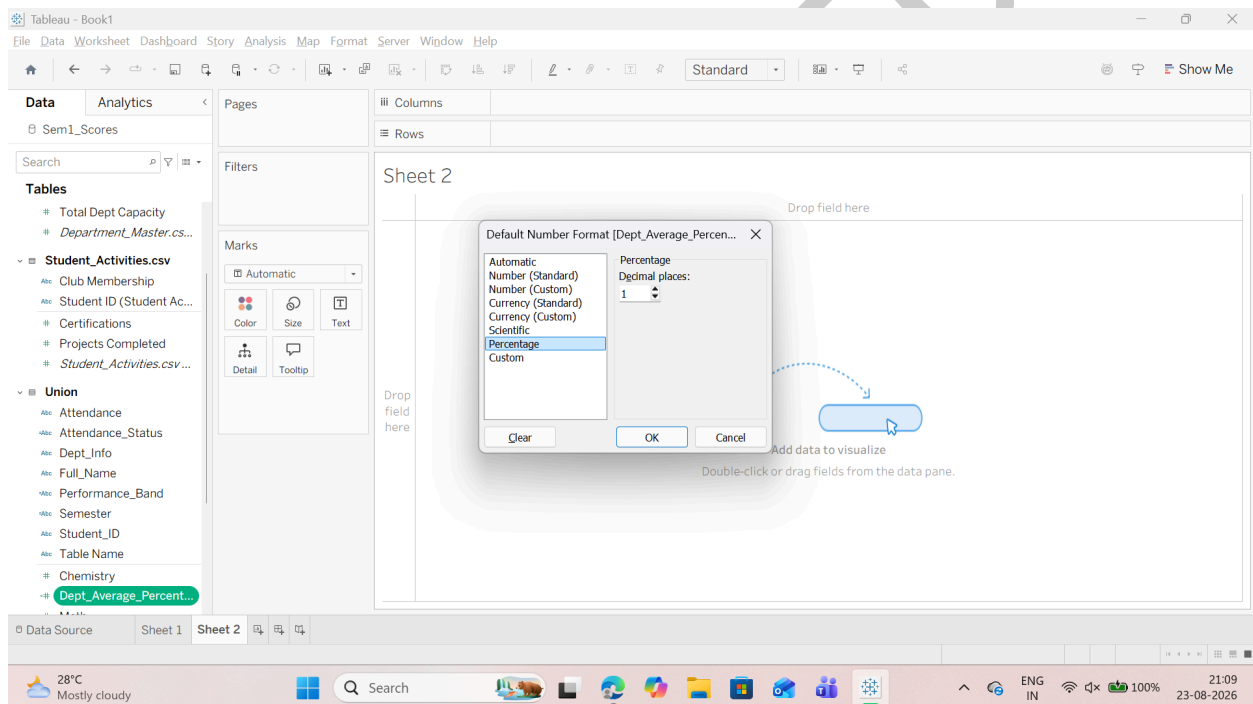
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F: Handle Different Levels of Detail in Datasets

Step 1: Compute Department-Level Benchmark using {FIXED} LOD

1. In Tableau, navigate to Sheet 1 (or create a new worksheet named Sheet 2).
2. In the Data Pane on the left, right-click an empty space --> select Create Calculated Field....
3. Create the department-wide benchmark field:
 - o Field Name: Dept_Average_Percentage_LOD
 - o Tableau Formula:
{ FIXED [Dept_Info] : AVG([Overall_Percentage]) }
4. Set default formatting:
 - o Right-click Dept_Average_Percentage_LOD --> Default Properties --> Number Format... --> select Percentage (1 decimal place) --> OK.

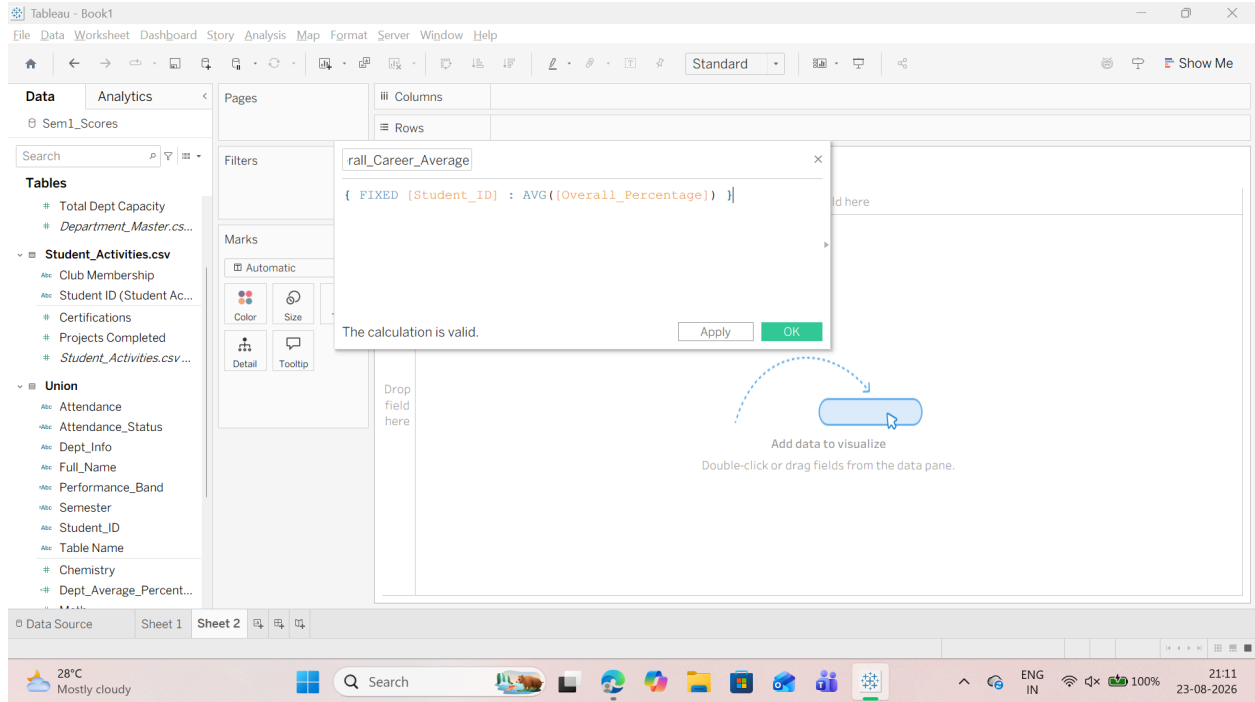


Step 2: Compute Student Multi-Semester Average using {FIXED} LOD

1. Right-click in the Data Pane --> select Create Calculated Field....
2. Create the cumulative student benchmark across all semesters:
 - o Field Name: Student_Overall_Career_Average
 - o Tableau Formula:
{ FIXED [Student_ID] : AVG([Overall_Percentage]) }
3. Set its Number Format to Percentage (1 decimal place, Just like we did in previous step).

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Step 3: Calculate Performance Variance (Student vs. Department Benchmark)

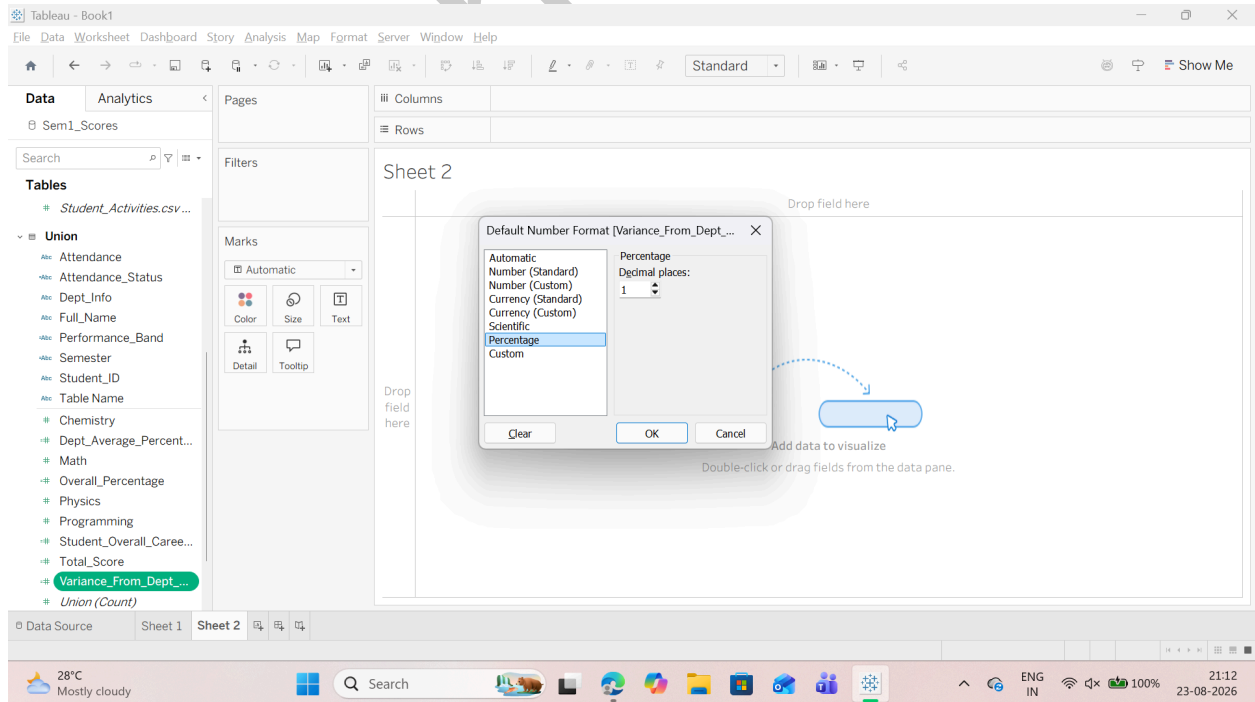
1. Right-click in the Data Pane --> select Create Calculated Field....

2. Create the relative difference metric:

o Field Name: Variance_From_Dept_Average

o Tableau Formula: $[Overall_Percentage] - [Dept_Average_Percentage_LOD]$

3. Set its Number Format to Percentage (1 decimal place).



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Step 4: Create a Benchmark Indicator Dimension

1. Right-click in the Data Pane -->select Create Calculated Field....

2. Create an automated comparison label:

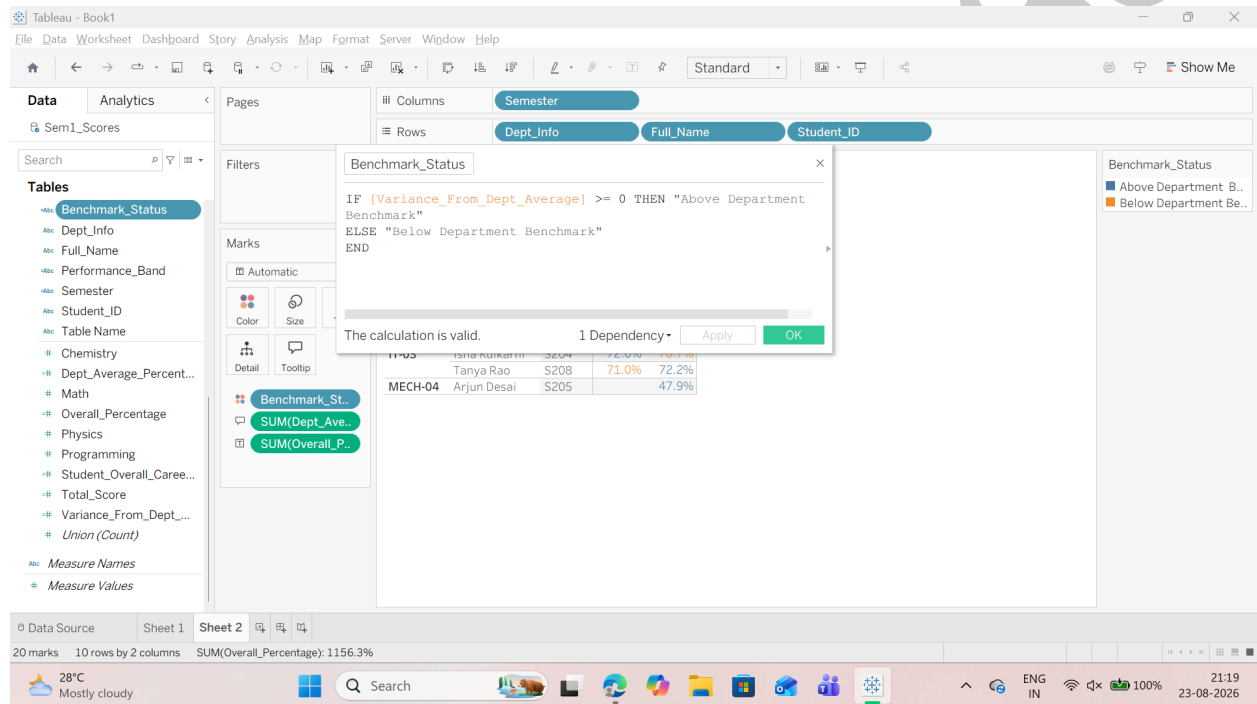
o Field Name: Benchmark_Status

o Tableau Formula:

IF [Variance_From_Dept_Average] >= 0 THEN "Above Department Benchmark"

ELSE "Below Department Benchmark"

END



Step 5: Build the LOD Benchmark Visualization

1. Drag Dept_Info, Student_ID, and Full_Name to the Rows shelf.

2. Drag Semester to the Columns shelf.

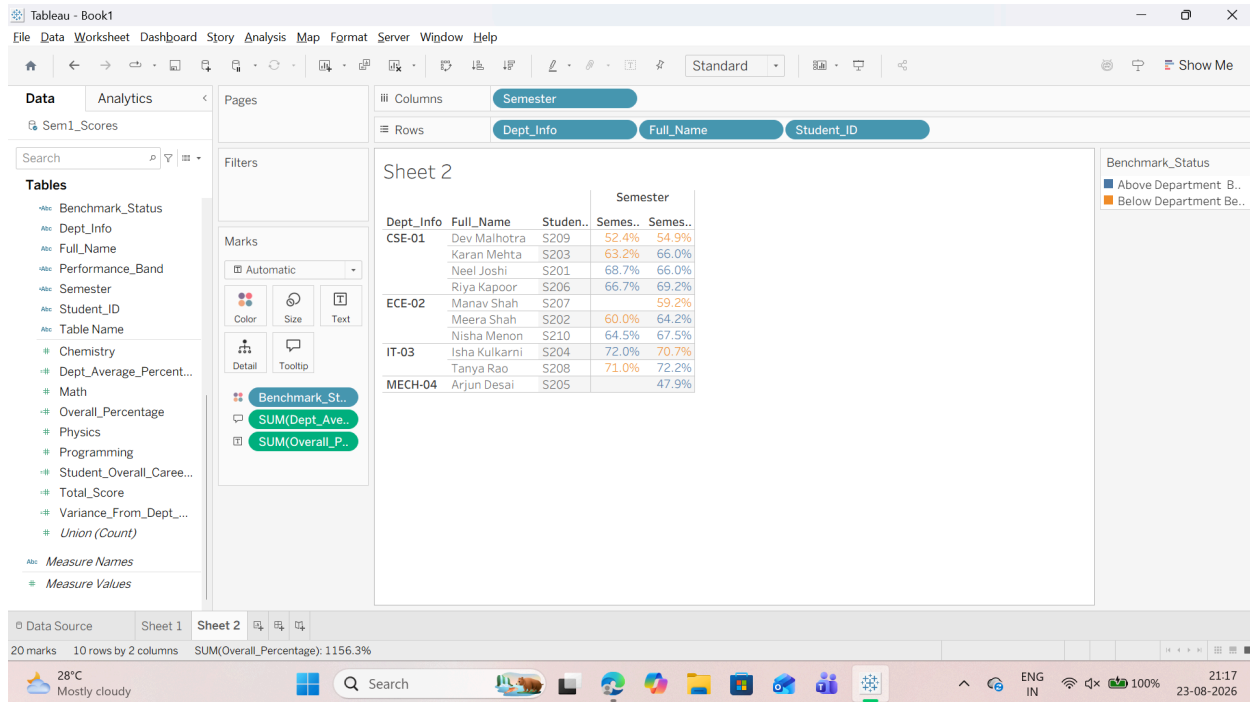
3. Drag Overall_Percentage onto Text on the Marks card.

4. Drag Dept_Average_Percentage_LOD onto Tooltip on the Marks card.

5. Drag Benchmark_Status onto Color on the Marks card.

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G: Apply Advanced Data Cleaning Techniques

Step 1: Clean Leading/Trailing Whitespace and Inconsistent Name Casing

1. Navigate to Sheet 1 (or a new worksheet).
2. Right-click an empty space in the Data Pane --> select Create Calculated Field....
3. Create the trimmed and standardized student name field:

o Field Name: Full_Name_Clean

o Tableau Formula:

```
UPPER(LEFT(TRIM([Full_Name]), 1)) +  
LOWER(MID(TRIM([Full_Name]), 2, FIND(TRIM([Full_Name]), " ") -  
1)) +  
" " +  
UPPER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 1,  
1)) +  
LOWER(MID(TRIM([Full_Name]), FIND(TRIM([Full_Name]), " ") + 2,  
LEN(TRIM([Full_Name]))))
```

o (Alternatively, for standard trim-only cleaning):

Code snippet

```
TRIM([Full_Name])
```

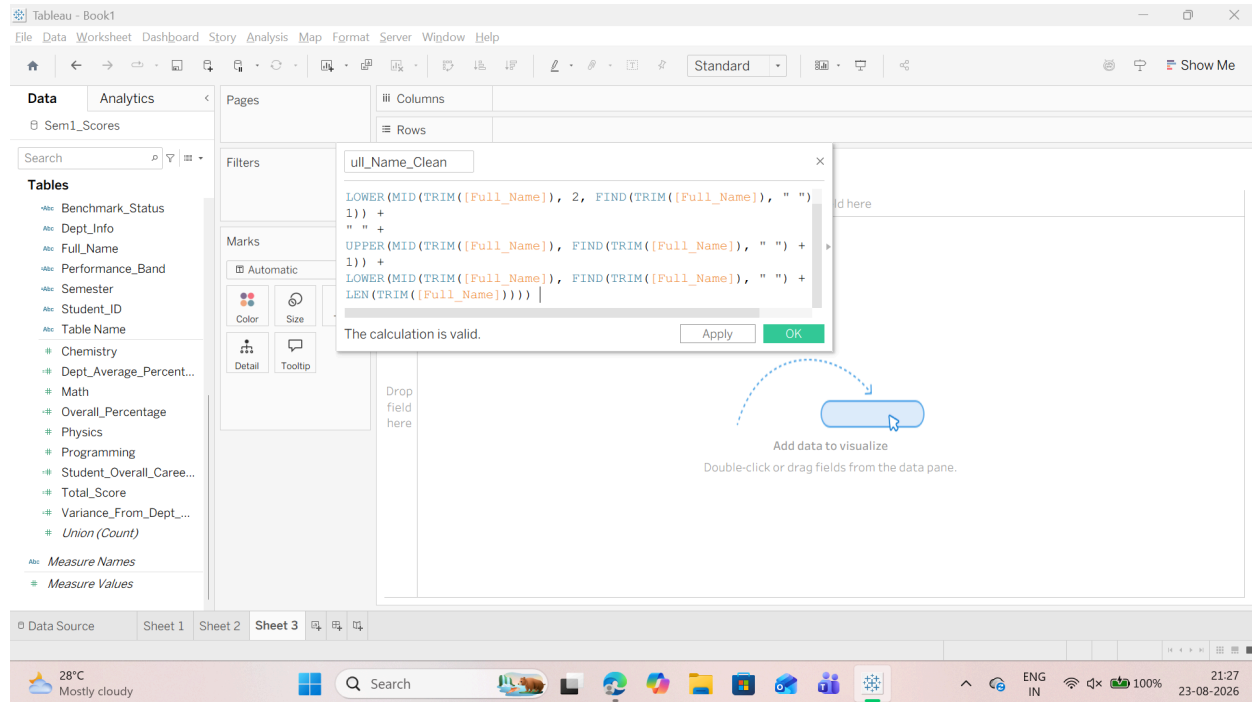
o Click OK.

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Step 2: Parse First Name and Last Name using SPLIT()

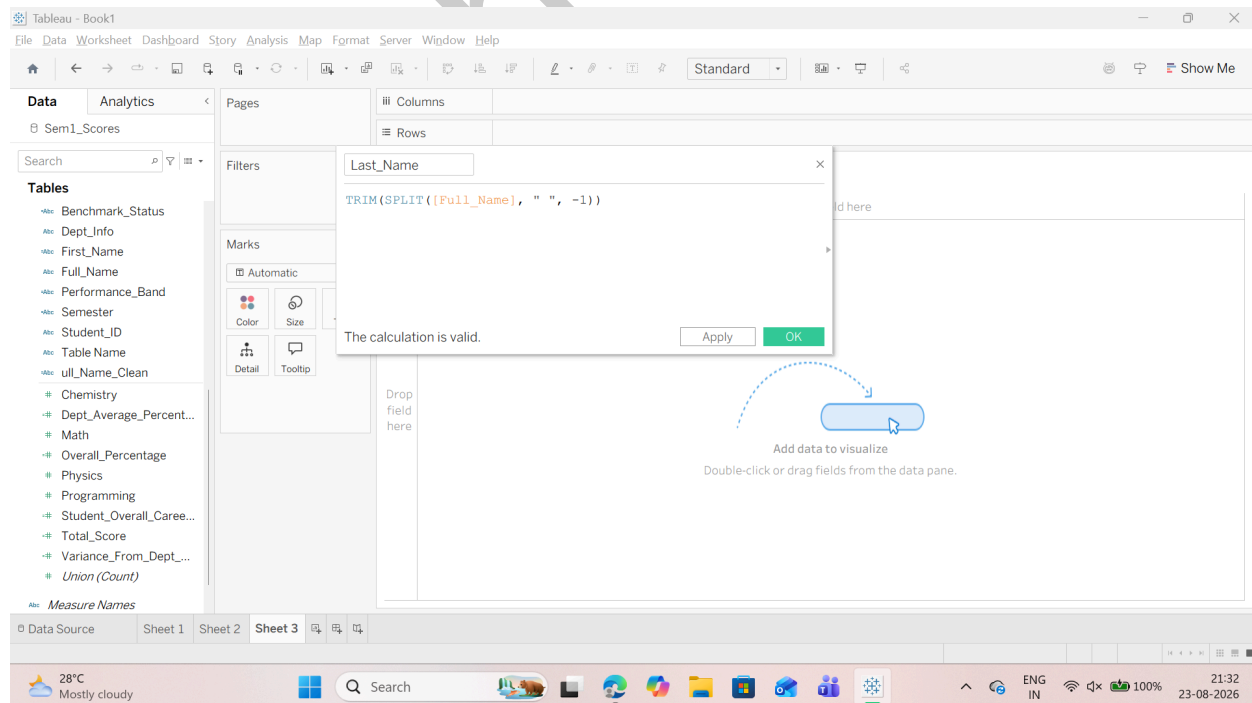
1. Extract the First Name:

o Right-click in the Data Pane --> select Create Calculated Field....

o Field Name: First_Name o Tableau Formula: `TRIM(SPLIT([Full_Name], " ", 1))` .

2. Extract the Last Name:

o Right-click in the Data Pane --> select Create Calculated Field.... o Field Name: Last_Name o Tableau Formula: `TRIM(SPLIT([Full_Name], " ", -1))`



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Step 3: Split Composite Department Codes (Dept_Info)

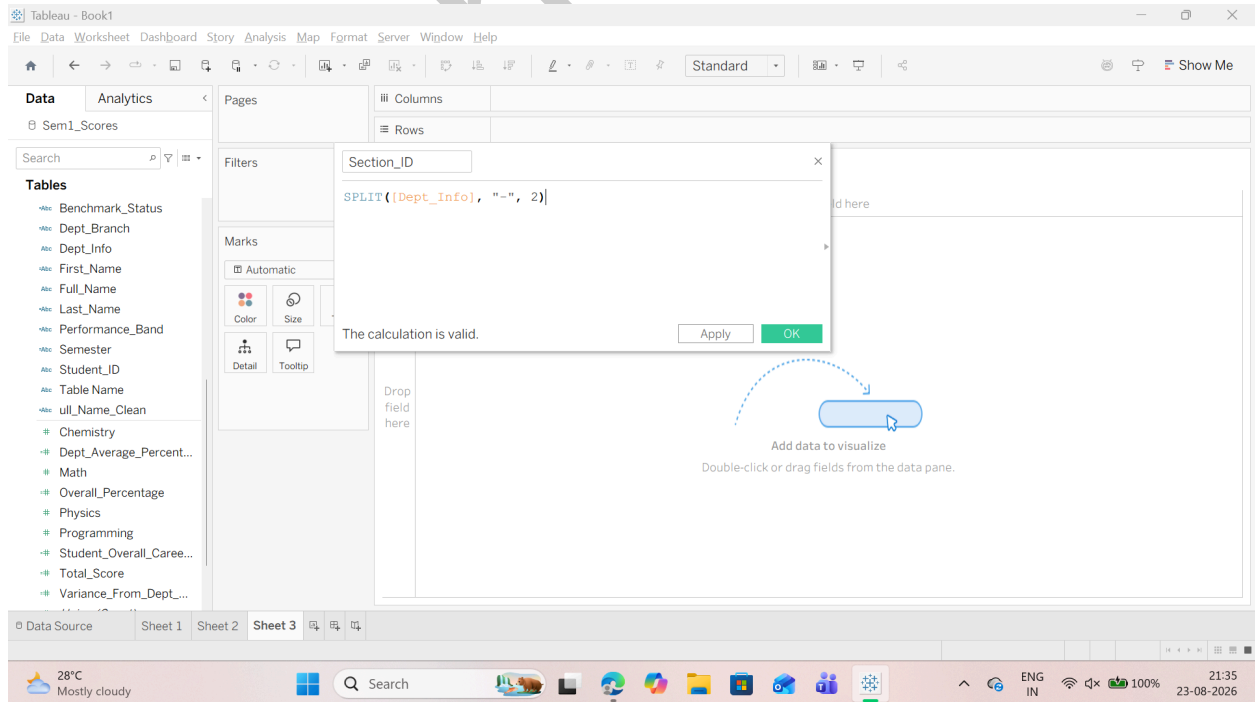
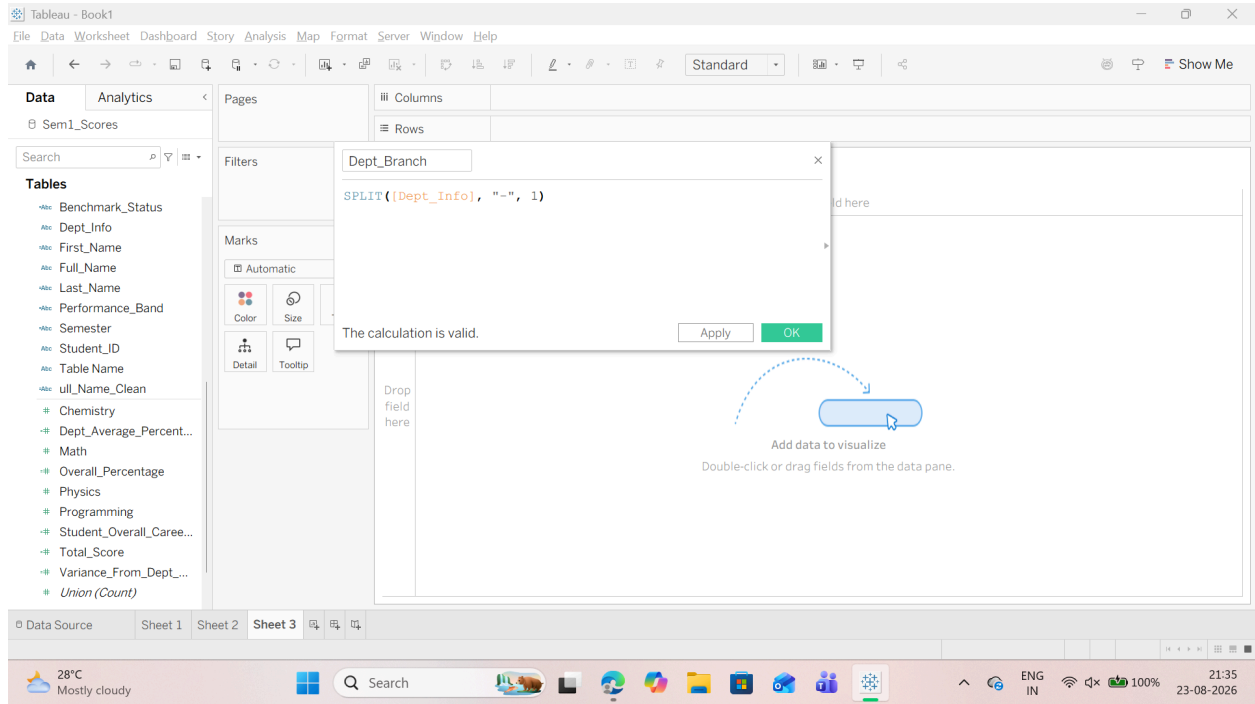
1. Extract the Department Acronym: 2. Extract the Batch / Section ID:

o Right-click in the Data Pane --> select Create Calculated Field....

o Field Name: Dept_Branch o Tableau Formula: SPLIT([Dept_Info], "-", 1)

o Right-click in the Data Pane --> select Create Calculated Field....

o Field Name: Section_ID o Tableau Formula: SPLIT([Dept_Info], "-", 2)



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Step 4: Create a Clean Composite Student Label

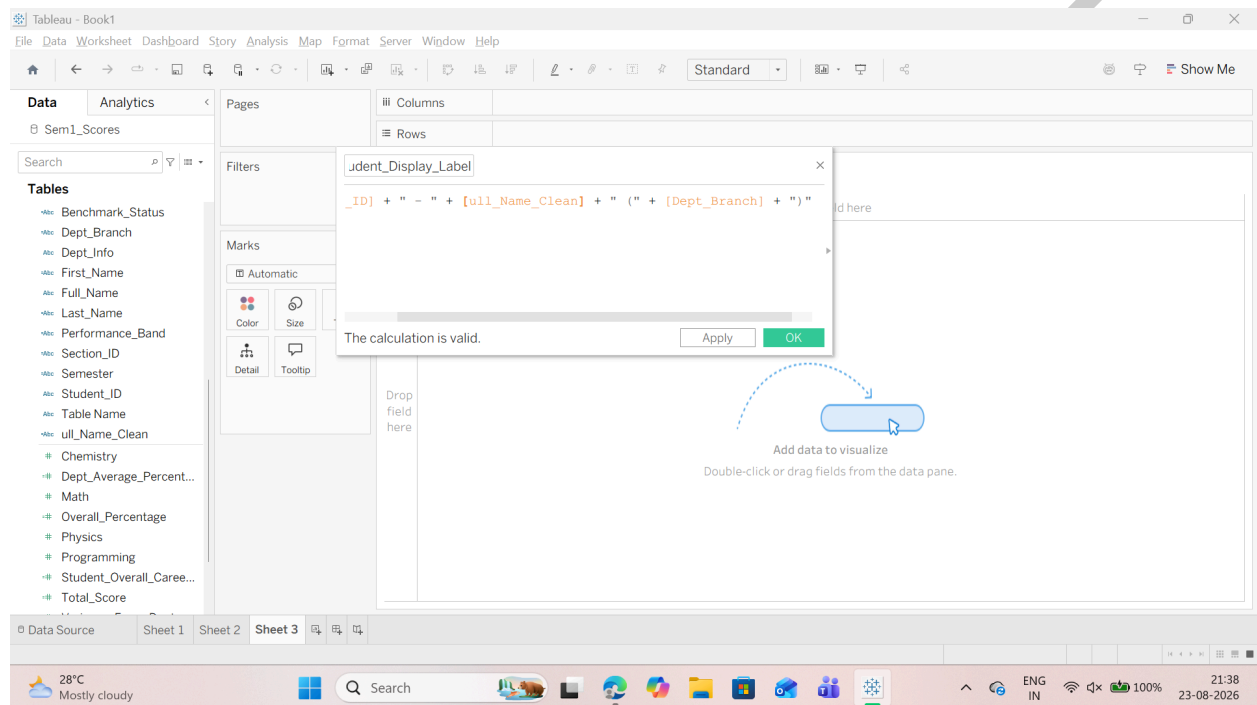
Combine cleaned names, student IDs, and department codes for clean dashboard titles and search filters:

1. Right-click in the Data Pane --> select Create Calculated Field....

2. Field Name: Student_Display_Label

3. Tableau Formula:

[Student_ID] + " - " + [Full_Name_Clean] + " (" + [Dept_Branch] + ")"

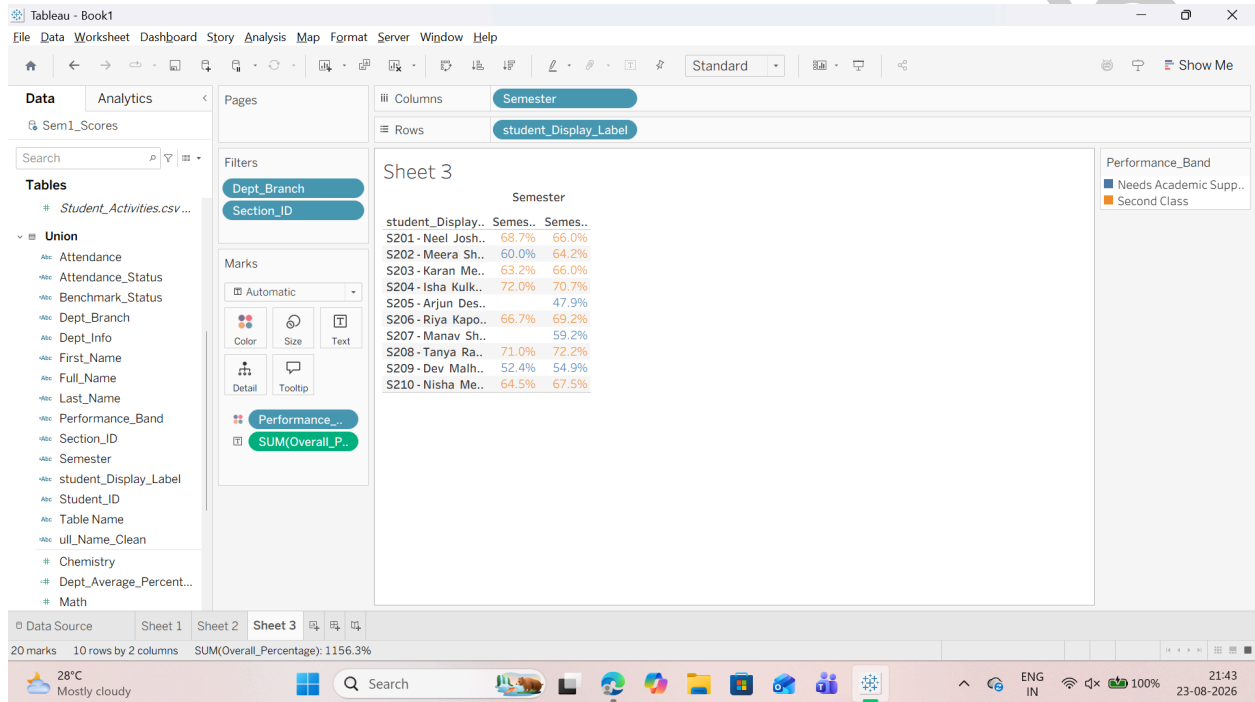


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Step 5: Verify in the Final Dashboard View

1. Place Student_Display_Label on the Rows shelf.
2. Place Semester on the Columns shelf.
3. Place Overall_Percentage on Text.
4. Place Performance_Band on Color.
5. Place Dept_Branch and Section_ID in the Filters shelf. (While Filtering You Can select any of the value you want)



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Shubham T085